

own country. The letter was intercepted and was brought to the notice of President Thomas Jefferson. Burr was acquitted after a trial but sadly, his reputation was tarnished forever.

Fascinated by cryptography, President Thomas Jefferson then went on to create a cipher of his own. However, it must be noted that a similar cipher was used by the US Navy a few years before the Jefferson cipher was invented.

Jefferson's Wheel

In 1795, Thomas Jefferson invented the 'wheel cipher', though he didn't use it much. It basically consisted of a set of wheels where the letters of the alphabet were randomly distributed. The key is the order of the wheels on the axle. Thus, the message could be coded by arranging the letters along the axis of the axle so that the messages are formed. Any other row of such letters can also be employed as ciphertext. But to decode, the recipient must align the ciphertext to the rotational axis. Without the knowledge of the arrangement of symbols on the wheels, a plaintext of any length can be churned out. This makes the cipher a secure one for the first use. But using the same wheels in the same order for multiple occasions can be used for statistical relays.

Developing on the wheel cipher, Colonel Decius Wadsworth created a set of two disks, where one was embedded inside the other, way back in 1817. The outer disk had the letters of the alphabet and numbers 2-8

while the inside disk featured only the alphabets. The disks were put together at a ratio of 26:33. To code a message, the inner disk is turned until the required letter is at the top position along with the number of turns required to result in the ciphertext. The ratio ensures that the repetition doesn't happen till all the 33 characters of the plaintext have been used. Sadly, Wadsworth wasn't credited for this design because Charles Wheatstone devised a rather similar machine and took away the limelight.



WWII Enigma Cipher Machine